

Network Security - Individual Assignment 2025

Secure by design

Assignment Overview

Task: For this individual MSc Cyber Security assignment, students are required to design and implement a secure network architecture using GNS3 network simulator. The network must incorporate a Cloud as a bridge to a VMWare network interface, enabling end-to-end connectivity and security testing with a Kali Linux virtual machine. Students should demonstrate appropriate security controls, justify their design decisions, and ensure the final topology allows practical validation of the network's resilience through the Kali VM.

Deadline: 19 December 2025

Submission Format: GNS3 save file, Technical Report (PDF).

Assignment Scenario

You are a security consultant hired by **TechCorp Ltd**, a medium-sized technology company with 200 employees. The company has experienced recent security incidents and requires a redesigned network infrastructure with robust security controls.

Business Requirements

- Support for three departments (**can be represented with less nodes to reduce resource use of the simulation, but the configuration should support this number of devices**):
 - Sales (40 users),
 - Development (80 users),
 - Management (20 users)
- Secure remote access for traveling employees
- DMZ hosting: Web server, Mail server, DNS server
- Guest WiFi network (isolated from corporate network)

Network Design & Implementation

Task 1.1: Network Architecture Design

Design a secure network topology that includes:

Core Components:

- Hierarchical network design (Core, Distribution, Access layers)

- Appropriate VLAN segmentation for departments
- DMZ for public-facing services
- Perimeter security with firewall/router
- Internal firewall between segments

IP Addressing Scheme:

- Use private IP ranges (RFC 1918)
- Implement subnetting for each department
- Reserve IP ranges for future growth
- Document IP allocation in your report

Required Devices:

- Multi-layer switches (minimum 2)
- Routers with security features
- Firewall/ASA device
- Servers (Web, DNS, DHCP, Email)
- End-user devices (PCs, laptops)
- Wireless access points
- Cloud devices for bridging into network segments for security evaluation

Task 1.2: Security Configuration

Implement the following security controls:

Access Control:

- Port security, MAC filtering, ACLs - Standard and essential
- VLAN access policies - Core networking security

Network Segmentation:

- VLANs with inter-VLAN routing – Fundamental security practice
- DMZ configuration - Essential for any security design
- Guest network isolation - Standard practice
- Management VLAN - Best practice

Additional Security:

- DHCP snooping and Dynamic ARP inspection - Important Layer 2 security
- Disable Telnet (SSH only) - Basic hardening
- Strong passwords - Obvious requirement
- Disable unnecessary services - Good practice

Minimum Requirements

You will need a registered GNS3 account (you will be prompted to get one when downloading/starting GNS3).

GNS:

- GNS3
- At least 3 network segments
- 1 firewall device
- Cloud node configured (bridge between the simulated network and Kali VM)

Testing Environment:

- Windows 11 host laptop
- GNS3 VM
- Kali Linux VM

Academic Integrity

This is an **individual assignment**. All design work, configurations, testing, and report writing must be your own. Plagiarism detection software will be used. Sharing Packet Tracer files or copying configurations constitutes academic misconduct.

Report requirements

1. **Executive summary** (200-300 words)
2. **Network Architecture design:** Design Rationale, Network Topology, IP address schemes
3. **Security control implementation:** Access Control Mechanisms, Network Segmentation, Additional Security Hardening
4. **Security Testing and Validation**
5. **Critical Evaluation:** strengths and potential limitations
6. **Conclusion** (200-500 words)
7. **References**